

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12407350

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Chinthu; Siva Kumar et al.

Voltage level shifter with programmable high supply voltage and high supply voltage-dependent variable low supply voltage

Abstract

A disclosed circuit structure includes a voltage level shifter connected to a variable voltage generator for receiving a low supply voltage and to a programmable voltage generator for receiving a high supply voltage that is higher than the low supply voltage. The high supply voltage is programmable to one of multiple different voltage levels. The variable voltage generator tracks the voltage level of the high supply voltage and outputs the low supply voltage based thereon. The high supply voltage is tracked either directly or using the trim bit signal employed by the programmable voltage generator to program the high supply voltage. The low supply voltage is high when the high supply voltage is high to ensure operation within the safe operating area and is low when the high supply voltage drops below some threshold level to ensure operability and, optionally, improve operating speed.

Inventors: Chinthu; Siva Kumar (Bangalore, IN), Gopinath; Venkatesh Periyapatna (Fremont, CA), Pasupula; Suresh (Bangalore, IN), Dwivedi; Devesh (Bangalore, IN)

Applicant: GlobalFoundries U.S. Inc. (Malta, NY)

Family ID: 94210697

Assignee: GlobalFoundries U.S. Inc. (Malta, NY)

Appl. No.: 18/350305

Filed: July 11, 2023

Prior Publication Data

Document Identifier

US 20250023564 A1

Publication Date

Jan. 16, 2025

Publication Classification

Int. Cl.: H03K19/0175 (20060101); H03K19/00 (20060101); H03K19/003 (20060101)

U.S. Cl.:

CPC H03K19/017581 (20130101); H03K19/0013 (20130101); H03K19/0016 (20130101); H03K19/00384 (20130101);

Field of Classification Search

CPC: H03K (19/0013); H03K (19/0016); H03K (19/00384); H03K (19/017581); H03K (19/018507); H03K (19/018521); H03K (19/018585); H03K (19/20)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4149232	12/1978	Eaton, Jr.	N/A	N/A
5059815	12/1990	Bill et al.	N/A	N/A
5436587	12/1994	Cernea	N/A	N/A
5734286	12/1997	Takeyama et al.	N/A	N/A
6028473	12/1999	Kamei et al.	N/A	N/A
6046626	12/1999	Saeki et al.	N/A	N/A
6046627	12/1999	Itoh	327/544	G11C 5/146
6099100	12/1999	Lee	N/A	N/A
6160723	12/1999	Liu	N/A	N/A
6191963	12/2000	McPartland et al.	N/A	N/A
6356469	12/2001	Roohparvar et al.	N/A	N/A
6392905	12/2001	Huang et al.	N/A	N/A
6483728	12/2001	Johnson et al.	N/A	N/A
6525949	12/2002	Johnson et al.	N/A	N/A
6570434	12/2002	Hsu et al.	N/A	N/A
6747897	12/2003	Karaki	N/A	N/A
6819162	12/2003	Pelliconi	N/A	N/A
6922096	12/2004	Cernea	N/A	N/A
6995603	12/2005	Chen et al.	N/A	N/A
7116156	12/2005	Myono et al.	N/A	N/A
7710183	12/2009	Chaba et al.	N/A	N/A
8049553	12/2010	Kim et al.	N/A	N/A
8476963	12/2012	Cook et al.	N/A	N/A
8963624	12/2014	Murakami	N/A	N/A
9013229	12/2014	Rahman et al.	N/A	N/A
9160930	12/2014	Ishii	N/A	N/A
9306450	12/2015	Emira et al.	N/A	N/A
9548655	12/2016	Ersoy et al.	N/A	N/A
10355694	12/2018	Kumar	N/A	N/A
10498230	12/2018	Chen et al.	N/A	N/A
10911047	12/2020	Shetty et al.	N/A	N/A

11139004	12/2020	Sinha et al.	N/A	N/A
11277121	12/2021	Goyal et al.	N/A	N/A
11569738	12/2022	Chinthu	N/A	N/A
11569838	12/2022	Chinthu	N/A	N/A
11671080	12/2022	Ross et al.	N/A	N/A
11843374	12/2022	Lee	N/A	N/A
2008/0012627	12/2007	Kato	N/A	N/A
2009/0134929	12/2008	Luo	327/333	H03K 3/012
2010/0309716	12/2009	Tsukada	N/A	N/A
2010/0327959	12/2009	Lee	N/A	N/A
2014/0132329	12/2013	Li et al.	N/A	N/A
2019/0379365	12/2018	Hsu et al.	N/A	N/A
2025/0023564	12/2024	Chinthu	N/A	H03K 19/00384
2025/0023566	12/2024	Chinthu	N/A	H03K 19/20

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
4072015	12/2021	EP	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 18/350,316, Office Action dated Oct. 9, 2024, 12 pages. cited by applicant

U.S. Appl. No. 18/350,327, Notice of Allowance dated Jan. 27, 2025, 8 pages. cited by applicant

Ballo et al., “A Review of Charge Pump Topologies for the Power Management of IoT Nodes,” MDPI Electronics, 8, 480, 2019, pp. 1-15. cited by applicant

Cavalheiro et al., “Insights into Tunnel FET-Based Charge-Pumps and Rectifiers for Energy Harvesting Applications,” IEEE Transactions on Very Large Scale Integration (VLSI) Systems, vol. 25, Issue 3, 2017, pp. 1-11. cited by applicant

Chen et al., “A Sub-100mV Ultra-Low Voltage Level-Shifter Using Current Limiting Cross-Coupled Technique for Wide-Range Conversion to I/O Voltage,” IEEE Access, vol. 8, published Aug. 6, 2020, pp. 145577-145585. cited by applicant

Lanuzza et al., “Low-Power Level Shifter for Multi-Supply Voltage Designs,” IEEE Transactions on Circuits and Systems—II: Express Briefs, vol. 59, No. 12, Dec. 2012, pp. 922-926. cited by applicant

Pan et al., “A Low Voltage to High Voltage Level Shifter Circuit for MEMS Application,” IEEE, In Proceedings of the 15th Biennial University/Government/Industry Microelectronics Symposium, Jul. 2003, pp. 128-131. cited by applicant

Pelliconi et al., “Power Efficient Charge Pump in Deep Submicron Standard CMOS Technology,” IEEE Journal of Solid-State Circuits, vol. 38, No. 6, Jun. 2003, pp. 1068-1071. cited by applicant

U.S. Appl. No. 17/488,380, Office Action dated Apr. 28, 2022, 11 pages. cited by applicant

U.S. Appl. No. 17/488,380, Response to Office Action submitted May 27, 2022, 19 pages. cited by applicant

U.S. Appl. No. 17/488,380, Final Office Action dated Aug. 31, 2022, 13 pages. cited by applicant

U.S. Appl. No. 17/488,380, Response to Final Office Action submitted Sep. 26, 2022, 21 pages. cited by applicant

U.S. Appl. No. 17/488,380, 2nd After Final Response dated Oct. 5, 2022, 10 pages. cited by applicant

U.S. Appl. No. 17/488,380, Advisory Action dated Oct. 5, 2022, 17 pages. cited by applicant

U.S. Appl. No. 17/488,380, Notice of Allowance dated Oct. 21, 2022, 9 pages. cited by applicant

Extended European Search Report for Related EP Application No. 24151286.2-1201 dated Jun. 14, 2024, 11 pages. cited by applicant

U.S. Appl. No. 18/350,316, Final Office Action dated Mar. 10, 2025, 13 pages. cited by applicant
U.S. Appl. No. 18/350,294, filed Jul. 11, 2023, 38 pages. cited by applicant
U.S. Appl. No. 18/350,316, filed Jul. 11, 2023, 47 pages. cited by applicant
U.S. Appl. No. 18/350,327, filed Jul. 11, 2023, 59 pages. cited by applicant
U.S. Appl. No. 18/350,316, Amendment to Office Action filed Dec. 9, 2024, 15 pages. cited by applicant
U.S. Appl. No. 18/350,316, Notice of Allowance dated Apr. 4, 2025, 9 pages. cited by applicant
U.S. Appl. No. 18/350,294, Office Action dated May 6, 2025, 13 pages. cited by applicant
U.S. Appl. No. 18/350,294, Response to Office Action filed Jul. 3, 2025, GF2022323-US-NP, 15 pages. cited by applicant

Primary Examiner: Crawford; Jason M

Attorney, Agent or Firm: Hoffman Warnick LLC

Background/Summary

BACKGROUND

(1) The present disclosure relates to on-chip voltage level shifters and, more particularly, to embodiments of a voltage level shifter with a programmable high positive supply voltage.

(2) Considerations in modern integrated circuit (IC) design include, but are not limited to, performance improvement, size scaling, and power consumption. Oftentimes design changes to improve one aspect of the design can yield an undesirable trade-off with respect to other aspects of the design. For example, a voltage level shifter in an IC design may be configured to translate (i.e., shift) a voltage signal from one logic level or voltage domain to another. Voltage level shifter designs have been developed that include low voltage transistors to reduce power consumption. However, at least some of the low voltage transistors in these voltage level shifters tend to quickly operate outside the safe operating area (SOA). Design modifications intended to avoid operation outside the SOA have included, for example, the use of asymmetric high voltage transistors (e.g., laterally diffused metal oxide semiconductor field effect transistors (LDMOSFETs)) in place of at-risk low voltage transistors and/or the use of a discrete biasing circuit. While such design modifications may help to avoid operation outside the SOA, they also add to circuit complexity, power consumption, and area.

SUMMARY

(3) Disclosed herein are embodiments of a structure. The structure includes a voltage level shifter with a low supply voltage node and a high supply voltage node. The structure further includes a programmable voltage generator that outputs a high supply voltage to the high supply voltage node and a variable voltage generator that outputs a low supply voltage to the low supply voltage node. The low supply voltage output by the variable voltage generator can specifically be dependent on and lower than the high supply voltage.

(4) In some embodiments, the structure can include a voltage level shifter with a low supply voltage node and a high supply voltage node. The structure can further include a programmable voltage generator that receives a trim bit signal and, based on the trim bit signal, outputs a high supply voltage to the high supply voltage node. The structure can further include a variable voltage generator that outputs a low supply voltage to the low supply voltage node. In order to set the voltage level of the low supply voltage, the variable voltage generator can also receive the trim bit signal and, based on the trim bit signal, can generate and output the low supply voltage such that it is lower than the high supply voltage.

(5) In other embodiments, the structure can include a voltage level shifter with a low supply voltage node and a high supply voltage node. The structure can further include a programmable voltage generator that outputs a high supply voltage to the high supply voltage node. The structure can also include a variable voltage generator that outputs a low supply voltage to the low supply voltage node. In order to set the voltage level of the low supply voltage, the variable voltage generator can receive the high supply voltage, internally generate a trim bit signal based on the high supply voltage, and further generate and output the low supply voltage based on the trim bit signal so it is lower than the high supply voltage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure will be better understood from the following detailed description with reference to the drawings, which are not necessarily drawn to scale and in which:
- (2) FIG. 1 is a circuit diagram illustrating embodiments of a disclosed structure;
- (3) FIG. 2 is a circuit diagram illustrating an example of a variable voltage generator that can be incorporated into the structure of FIG. 1;
- (4) FIG. 3 is a circuit diagram illustrating another example of a variable voltage generator that can be incorporated into the structure of FIG. 1;
- (5) FIG. 4 is a circuit diagram illustrating an example of a voltage divider that can be incorporated into the variable voltage generator of FIG. 3; and
- (6) FIG. 5 is a timing diagram illustrating input voltage pulses to and output voltage pulses from the structure of FIG. 1.

DETAILED DESCRIPTION

(7) As mentioned above, considerations in modern integrated circuit (IC) design include, but are not limited to, performance improvement, size scaling, and power consumption. Oftentimes design changes to improve one aspect of the design can yield an undesirable trade-off with respect to other aspects of the design. For example, a voltage level shifter in an IC design may be configured to translate (i.e., shift) a voltage signal from one logic level or voltage domain to another. Voltage level shifter designs have been developed that include low voltage transistors to reduce power consumption. However, at least some of the low voltage transistors in these voltage level shifters tend to quickly operate outside the safe operating area (SOA). Design modifications intended to avoid operation outside the SOA have included, for example, the use of asymmetric high voltage transistors (e.g., laterally diffused metal oxide semiconductor field effect transistors (LDMOSFETs)) in place of at-risk low voltage transistors and/or the use of a discrete biasing circuit. While such design modifications may help to avoid operation outside the SOA, they also add to circuit complexity, power consumption, and area.

(8) In view of the foregoing, disclosed herein are embodiments of a circuit structure that includes a voltage level shifter, a variable voltage generator, and a programmable voltage generator. The voltage level shifter can include a low supply voltage node, a high supply voltage node, and parallel branches, including multiple transistors, connected between the low and high supply voltage nodes. The variable voltage generator can generate a variable low supply voltage (VL) and can apply VL to the low supply voltage node. The programmable voltage generator can generate a programmable high supply voltage (VH) and can apply VH to the high supply voltage node. The VH can be any of multiple (e.g., up to three) different VH levels (e.g., a relatively low first VH level of 1.8 volts (V), a relatively high second VH level of 3.6V, and a mid-level third VH level between the first and second VH levels, such as at 2.6V). The variable voltage generator can be configured to track the voltage level of VH and to adjust the voltage level of VL based thereon. In some embodiments, such tracking can be performed using a trim bit signal employed by the

programmable voltage generator (e.g., a charge pump thereof) when programming VH. In other embodiments such tracking can be achieved by direct monitoring of VH. In either case, VL can be lower than VH and the variable voltage generator can increase VL from a relatively low first VL level (e.g., of 0.8V) to a relatively high second VL level (e.g., of 1.8 volts (V)) in response to a relatively high second VH (e.g., of 3.6V) to ensure operation within the safe operating area (SOA). The variable voltage generator can further decrease VL from the second VL level (e.g., of 1.8V) back to the first VL level (e.g., of 0.8V) when VH drops to the first VH level (e.g., of 1.8V) to ensure that the voltage level shifter remains operable and, optionally, even when VH drops to the mid-level third VH level (e.g., of 2.6V) to improve operating speed.

(9) FIG. 1 is a schematic diagram generally illustrating embodiments of a circuit structure **199** including a voltage level shifter **100**, a programmable voltage generator **155**, and a variable voltage generator **150**.

(10) The voltage level shifter **100** can, for example, include multiple transistors. As discussed in greater detail below, these transistors can include both P-type transistors and N-type transistors. For purposes of illustration, the P-type transistors and N-type transistors are described below and illustrated in the figures as being metal oxide semiconductor field effect transistors (MOSFETs) and, particularly, P-type field effect transistors (PFETs) and N-type field effect transistors (NFETs), respectively. A MOSFET refers to a transistor with a semiconductor channel region positioned laterally between a source region and a drain region and with a gate (e.g., including a gate dielectric-gate conductor stack) adjacent to the channel region. However, the figures and discussion thereof are not intended to be limiting. For example, alternatively, a similar circuit structure could be formed using bipolar junction transistors (BJTs) and, particularly, PNP BJTs and NPN BJTs. The transistors can, for example, all have the same maximum voltage rating (VDDW) and, for reduced power consumption, VDDW can be relatively low as specified by the transistor data sheet. For example, the transistors can have a maximum voltage rating of 3.3V or lower. In some embodiments, the transistors can have a maximum voltage rating of 1.8V, 1.5V or 0.8V. To minimize manufacturing complexity, the transistors can all be symmetric. That is, they can be designed so that the source region and the drain region are the same (e.g., same size, doping, etc.) except for minor process variations and so that the same maximum voltage rating applies to the gate-source voltage (VGS), the gate-drain voltage (VGD), and the drain-source voltage (VDS). As discussed in greater detail below, the voltage level shifter **100** can specifically be configured to achieve voltage level shifting without violating any of these maximum voltage ratings in order to avoid device stress and ensure operation within the SOA.

(11) For purposes of illustration, examples provided herein are generally based on all transistors having a 1.8V maximum voltage rating. Additionally, various components of the circuit structure can, as discussed in greater detail below, be connected to receive fixed supply voltages. The fixed supply voltages available to the circuit structure can include but are not limited to: a fixed supply voltage (VDDW), which is set at a voltage level equal to the maximum voltage rating of the transistors (e.g., VDDW=1.8V), and an additional fixed supply voltage (VDDC), which is set at a voltage level between ground and VDDW. For example, VDDC can be essentially $\frac{1}{2}$ the magnitude of VDDW, less than $\frac{1}{2}$ the magnitude of VDDW, etc. For example, if VDDW=1.8V, VDDC=0.8V.

(12) The programmable voltage generator **155** can be configured to generate and output a programmable high supply voltage (VH) and the variable voltage generator **150** can be configured to generate and output a variable VH-dependent low supply voltage (VL). The voltage level shifter **100** can include a high supply voltage node, which is electrically connected to the programmable voltage generator **155** for receiving VH, and a low supply voltage node **191**, which is electrically connected to the variable voltage generator **150** for receiving VL.

(13) More specifically, the programmable voltage generator **155** can be configured to output VH at a relatively low first VH level, a relatively high second VH level, and, optionally, at some medium-level third VH level (between the first and second VH levels). The programmable voltage generator

155 can be configured so that the second VH level is essentially double the magnitude of VDDW, so that the first VH level is equal to VDDW and, if applicable, so that the third VH level is somewhere between the first and second VH levels. For example, in some embodiments, VDDW is 1.8V, the first VH level is 1.8V, the second VH level is 3.6V, and, if applicable, the third VH level is 2.6V. For purposes of this disclosure, “essentially double” the magnitude refers to twice the magnitude, plus or minus 10% thereof (e.g., if VDDW is 1.8V, then the high VH=3.6V+/-0.36V, etc.). Programmable voltage generators configured to output different supply voltages are well known in the art and, thus, the details thereof have been omitted from this specification in order to allow the reader to focus on the salient aspects of the disclosed embodiments. However, those skilled in the art will recognize that, typically, a programmable voltage generator will include a charge pump configured to convert an input direct current (DC) power supply to one or more different DC power supplies and the state of a trim bit signal (TRIM) received by the programmable voltage generator will dictate the level of VH output to a particular supply voltage node (in this case to the high supply voltage node **192**). For example, for a two-VH level programmable voltage generator, a single-bit two-state TRIM can control whether a high VH or a low VH is output to the high supply voltage node **192**. When TRIM is at a high logic level (i.e., a “1” logic level or a logic 1), the high VH can be output to the second supply node **192** and, when TRIM is at a low logic level (i.e., a “0” logic level or logic 0), the low VH can be output to the second voltage node **192**. For a three-VH programmable voltage generator, a tri-state single-bit TRIM can control whether a high VH, a mid-level VH, or a low VH is output to the high supply voltage node **192**. When TRIM is at logic 1, the high VH can be output to the second supply node **192**. When TRIM is at logic 0, the mid-level VH can be output to the second voltage node **192**. When TRIM is at a high impedance (i.e., high Z) state, the low VH can be output to the second voltage node **192**.

(14) The variable voltage generator **150** can be configured to track the voltage level of VH and generate and output VL depending upon the voltage level of VH (i.e., to generate and output a VH-dependent VL) in order to, for example, ensure voltage level shifter transistor operation within the SOA when VH is high, to ensure voltage level shifter operability when VH is low and, if applicable, to improve operating speed when VH is at some mid-level. In some embodiments, such tracking can be performed by the variable voltage generator **150** using an external trim bit signal (TRIM) employed by the programmable voltage generator **155** when programming VH. In another embodiment such tracking can be achieved by direct monitoring of VH.

(15) FIG. 2 is a schematic diagram illustrating one example of a variable voltage generator **150A** that can be incorporated into circuit structure **199** of FIG. 1 for supplying a VH-dependent VL to the low supply voltage node **191** of the voltage level shifter. In this example, the variable voltage generator **150A** is configured to track VH using TRIM **202**, which is externally generated and also employed by the programmable voltage generator **155** to establish the voltage level of VH. The variable voltage generator **150A** is further configured to generate and output VL at a first VL level (e.g., at 0.8V) or at a higher second VL level (e.g., at 1.8V) depending upon the state of TRIM **202** as well as the state of an enable signal (EN) **201**.

(16) In the disclosed embodiments, the circuit structure **199** can, for example, be incorporated into a memory circuit, such as a resistive random access memory (RRAM) circuit. In this case, EN can be a write enable signal, which is high during a write mode, and which is low during a read mode. For example, when EN **201** is at logic 0 and TRIM **202** is at a high-Z state (e.g., indicating that the programmable voltage generator **155** is outputting VH at the low first VH level, such as at 1.8V) or when EN **201** is at logic 1 and TRIM is at logic 0 (e.g., indicating that the programmable voltage generator **155** is outputting VH at a medium third VH level, such as a 2.6V), the variable voltage generator **150A** can output VL at the low first VL level (e.g., at 0.8V). However, when EN **201** and TRIM **202** are both at logic 1 (e.g., indicating that the programmable voltage generator **155** is outputting VH at the high second VH level, such as at 3.6V), the variable voltage generator **150A**

can output VL at the high second VL level (e.g., at 1.8V).

(17) More specifically, the variable voltage generator **150A** can include: an inverter **260**; a pair of NAND gates (i.e., a first NAND gate **240** and a second NAND gate **250**); three PFETs (i.e., a first PFET **210**, a second PFET **220**, and a third PFET **230**); and an output voltage node **299**. The output voltage node **299** can be electrically connected to the low supply voltage node **191** of the voltage level shifter **100** for supplying VL to the low supply voltage node **191**. The first PFET **210** can include: a source region, which is electrically connected to receive VDDC; a drain region, which is electrically connected to the output voltage node **299**; and a gate, which is electrically connected to receive EN **201**. The second PFET **220** can include: a source region, which is electrically connected to receive VDDC; a drain region, which is electrically connected to the output voltage node **299**; and a gate, which is electrically connected to receive an output signal from the first NAND gate **240**. The third PFET **230** can include: a source region, which is connected to receive VDDW; a drain region, which is connected to the output voltage node **299**; and a gate, which is electrically connected to receive an output signal from the second NAND gate **250**. The inverter **260** can be electrically connected to receive, as an input, the externally generated TRIM **202** and can be configured to output an inverted trim bit signal (TRIMB) **203** (i.e., TRIMB **203** can be inverted with respect to TRIM **202**). The first NAND gate **240** can be electrically connected to receive, as inputs, EN **201** and TRIMB **203**, respectively, and can be configured to output a first NAND gate output signal (according to a conventional NAND gate truth table). The second NAND gate **250** can be electrically connected to receive, as inputs, EN **201** and TRIM **202**, respectively, and can be configured to output a second NAND gate output signal (according to a conventional NAND gate truth table).

(18) Those skilled in the art will recognize that a conventional NAND gate truth table specifies that when the inputs to a NAND gate are both low (i.e., 00) or are different (e.g., 01 or 10), then the NAND gate output signal will be high. When the inputs to the NAND gate are both at high (i.e., 11), then the NAND gate output signal will be low (i.e., 0).

(19) In operation, the on/off state of the first PFET **210** is controlled by EN **201**, the on/off state of the second PFET **220** is controlled by the output signal from the first NAND gate **240**, and the on/off state of the third PFET **230** is controlled by the output signal from the second NAND gate **250**. When EN is low, the PFET **210** will turn on. Additionally, the output signals from the first and second NAND gates **240** and **250** will both be high so the second PFET **220** and the third PFET **230** will be turned off. Thus, a VL at the low first VL level of VDDC (e.g., at 0.8V) will be supplied by the first PFET **210** to the output voltage node **299** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**. When EN **201** is high, TRIM **202** is low, and TRIMB **203** is high, the first PFET **210** will turn off, the output signal from the second NAND gate **250** will be high so the third PFET **230** will turn off, and the output signal from first NAND gate **240** will be low so the second PFET **220** will turn on. Thus, a VL at the low first VL level of VDDC is supplied through the second PFET **220** to the output voltage node **299** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**. Finally, when EN **201** is high, TRIM **202** is high, and TRIMB **203** is low, the first PFET **210** is off, the output signal from the first NAND gate **240** will be high so the second PFET **220** is off, and the output signal from the second NAND gate **250** will be low so the third PFET **230** turns on. Thus, a VL at the high second VL level of VDDW (e.g., at 1.8V) is supplied through the third PFET **230** to the output voltage node **299** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**.

(20) FIG. 3 is a schematic diagram illustrating another example of a variable voltage generator **150B** that can be incorporated into the circuit structure **199** of FIG. 1 for supplying a VH-dependent VL to the low supply voltage node **191** of the voltage level shifter **100**. In this example, the variable voltage generator **150B** is specifically configured to track VH through direct monitoring.

(21) Specifically, the variable voltage generator **150B** can include: a voltage divider **380**; a

comparator **385**; an inverter **360**; a pair of NAND gates (i.e., a first NAND gate **340** and a second NAND gate **350**); three PFETs (i.e., a first PFET **310**, a second PFET **320**, and a third PFET **330**); and an output voltage node **399**. The output voltage node **399** can be electrically connected to the low supply voltage node **191** of the voltage level shifter **100** for supplying the VL to the low supply voltage node **191**.

(22) Voltage divider **380** can be connected to receive V_H from the programmable voltage generator **155**. The voltage divider **380** can further be configured to divide V_H essentially in half (e.g., half+/-10%) and, more particularly, to output an output voltage that is essentially half the magnitude of V_H (i.e., $V_H/2$). FIG. 4 is a schematic diagram illustrating an example of a voltage divider **380** that can be incorporated into the variable voltage generator **150B**. As illustrated, this simple voltage divider can include a pair of resistors **401**, **402** connected in series between an input node for receiving V_H and ground. $V_H/2$ can be output at the junction between the two resistors **401**, **402**. The voltage divider shown in FIG. 4 is provided for illustration purposes. Alternatively, any other suitable voltage divider circuit could be employed to generate and output $V_H/2$ from V_H .

(23) Comparator **385** can be connected to receive, as inputs, $V_H/2$ and a reference voltage (V_{REF}), respectively. V_{REF} can be at some predetermined voltage level between the high second V_H level divided by two and the low first V_H level divided by two. For example, if the high second V_H level is 3.6V so that high $V_H/2$ is 1.8V and if the medium third V_H level at 2.6V so that the medium third V_H level divided by two is 1.3V, then V_{REF} can be equal to or greater than 1.3V but less than 1.8V. Comparator **385** can further be configured to compare $V_H/2$ to V_{REF} and to internally generate and output a trim bit signal (TRIM **302**) based on the results of the comparison. Specifically, comparator **385** can output TRIM **302** at a logic 0, when $V_H/2$ is less equal to or less than V_{REF} , whereas comparator **385** can output TRIM **302** at a logic 1, when $V_H/2$ is greater than V_{REF} .

(24) The remaining components of the variable voltage generator **150B** can be essentially the same as the components of the variable voltage generator **150A**, described above. Specifically, the first PFET **310** can include: a source region, which is electrically connected to receive V_{DDC} (e.g., at 0.8V); a drain region, which is electrically connected to the output voltage node **399**; and a gate, which is electrically connected to receive EN **301**. The second PFET **320** can include: a source region, which is electrically connected to receive V_{DDC} ; a drain region, which is electrically connected to the output voltage node **399**; and a gate, which is electrically connected to receive an output signal from the first NAND gate **340**. The third PFET **330** can include: a source region, which is connected to receive V_{DDW} ; a drain region, which is connected to the output voltage node **399**; and a gate, which is electrically connected to receive an output signal from the second NAND gate **350**. The inverter **360** can be electrically connected to receive, as an input, TRIM **302** (which in this case has been internally generated) and can be configured to output an inverted trim bit signal (TRIMB) **303** (i.e., TRIMB **303** is inverted with respect to TRIM **302**). The first NAND gate **340** can be electrically connected to receive, as inputs, EN **301** and TRIMB **303**, respectively, and can be configured to output a first NAND gate output signal (according to a conventional NAND gate truth table). The second NAND gate **350** can be electrically connected to receive, as inputs, EN **301** and TRIM **302**, respectively, and can be configured to output a second NAND gate output signal (according to a conventional NAND gate truth table).

(25) In operation, the on/off state of the first PFET **310** is controlled by EN **301**, the on/off state of the second PFET **320** is controlled by the output signal from the first NAND gate **340**, and the on/off state of the third PFET **330** is controlled by the output signal from the second NAND gate **350**. When EN **301** is low, the first PFET **310** turns on. Additionally, the output signals from the first and second NAND gates **340** and **350** will both be high so the second PFET **320** and the third PFET **330** will be off. Thus, a VL at the low first VL level of V_{DDC} (e.g., at 0.8V) will be supplied by the first PFET **310** to the output voltage node **399** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**. When EN **301** is high, TRIM **302** is low and TRIMB **303** is

high, the first PFET **310** is off the output signal from the second NAND gate **350** is high so the third PFET **330** is off, and the output signal from first NAND gate **340** is low so the second PFET **320** turns on. Thus, a VL at the low first VL level of VDDC (e.g., at 0.8V) is supplied through the second PFET **320** to the output voltage node **399** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**. Finally, when EN **301** is high, TRIM **302** is high, and TRIMB **303** is low, the first PFET **310** turns off, the output signal from the first NAND gate **340** is high so the second PFET **320** turns off, and the output signal from the second NAND gate **350** is low so the PFET **330** turns on. Thus, a VL at the high second VL level of VDDW (e.g., at 1.8V) is supplied through the third PFET **330** to the output voltage node **299** and, thereby to the low supply voltage node **191** of the voltage level shifter **100**.

(26) Referring again to FIG. **1**, the voltage level shifter **100** can include symmetric branches **101a-101b** connected in parallel between the low supply voltage node **191** and the high supply voltage node **192**.

(27) Each branch **101a, 101b** can include a first PFET **110a, 110b** and a second PFET **120a, 120b** connected in series between low supply voltage node **191** and high supply voltage node **192**. That is, first PFET **110a, 110b** can include a drain region **113a, 113b** electrically connected to low supply voltage node **191**, a gate **112a, 112b**, and a source region **111a, 111b**. Second PFET **120a, 120b** can include a drain region **123a, 123b** electrically connected to source region **111a, 111b** of first PFET **110a, 110b**, a gate **122a, 122b**, and a source region **121a, 121b** electrically connected to high supply voltage node **192**. In some embodiments, source region **111a, 111b** of first PFET **110a, 110b** and drain region **123a, 123b** of the second PFET **120a, 120b** can be a shared source/drain region. In other embodiments, source region **111a, 111b** of the first PFET **110a, 110b** and drain region **123a, 123b** of the second PFET **120a, 120b** could be discrete regions electrically connected by an interconnect.

(28) Each branch **101a, 101b** can further include an intermediate output node **125a, 125b** and a third PFET **130a, 130b** electrically connected to a junction **116a, 116b** between first PFET **110a, 110b** and second PFET **120a, 120b**. Third PFET **130a, 130b** can specifically include: a source region **131a, 131b**, which is electrically connected to junction **116a, 116b** between first PFET **110a, 110b** and second PFET **120a, 120b**; a drain region **133a, 133b**, which is electrically connected to a final output node **135a, 135b** and also to gate **112a, 112b** of first PFET **110a, 110b**, and a gate **132a, 132b**, which is electrically connected to, e.g., low supply voltage node **191**, so as to receive VL.

(29) Additionally, the second PFET **120a** of branch **101a** and second PFET **120b** of branch **101b** can be cross-coupled. That is, gate **122a** of second PFET **120a** in branch **101a** can be electrically connected to junction **116b** between first PFET **110b** and second PFET **120b** in branch **101b**, while gate **122b** of second PFET **120b** in branch **101b** can be electrically connected to junction **116a** between first PFET **110a** and second PFET **120a** in branch **101a**.

(30) Finally, each branch **101a, 101b** can further include an NFET **140a, 140b** (also referred to herein as a pass-gate NFET). NFET **140a, 140b** can include: a source region **141a, 141b**, which is electrically connected to an input node **145a, 145b**; a drain region **143a, 143b**, which is electrically connected to final output node **135a, 135b** (and thereby electrically connected to the drain region **133a, 133b** of third PFET **130a, 130b** as well as gate **112a, 112b** of first PFET **110a, 110b**); and a gate, which is electrically connected to receive a pass-gate control voltage that is, for example, equal to the maximum voltage rating (VDDW) of the transistors therein (e.g., 1.8V).

(31) The voltage level shifter **100** can further be connected to receive two input voltage pulses at the two input nodes **145a** and **145b**, respectively. Input voltage pulse (IN) at input node **145a** can transition between VDDW (e.g., at 1.8V) and ground (GND) (e.g., at 0.0V). An inverted input voltage pulse (INB) at input node **145b** can be inverted with respect to IN and, thus, can transition from GND to VDDW as IN transitions from VDDW to GND and vice versa.

(32) For purposes of this disclosure, a “voltage pulse” refers to an essentially rectangular direct current voltage signal that transitions between a low voltage level and a high voltage level at

regular and repeated intervals such that the pulse is at the low voltage level for a first time period (T1), switches to the high voltage level and remains at the high voltage level for a second time period (T2) (which is the same as or different from T1), switches back to the low voltage level and again remains at the low voltage level for T1, and so on.

(33) The voltage level shifter **100** can further be configured to generate and output four different output voltage pulses including two intermediate output voltage pulses at intermediate output nodes **125a-125b** and two final output voltage pulses at final output nodes **135a-135b**. The two intermediate output voltage pulses can include intermediate output voltage pulse (OUT.sub.i), which will transition between VL and VH, at intermediate output node **125b** and inverted intermediate output voltage pulse (OUTB.sub.i), which is inverted with respect to OUT.sub.i and, thus, will transition from VH to VL as OUT.sub.i transitions from VL to VH and vice versa, at intermediate output node **125a**. The two final output voltage pulses can include final output voltage pulse (OUT), which transitions between ground and VH, at a final output node **135b** and inverted final output voltage pulse (OUTB), which is inverted with respect to OUT and, thus, which transitions from VH to ground as OUT transitions from ground to VH and vice versa, at final output node **135a**.

(34) FIG. 5 shows an example of a timing diagram illustrating input voltage pulses (IN, INB) to a voltage level shifter **100** and output voltage pulses (OUT.sub.i, OUTB.sub.i, OUT, OUTB) from the voltage level shifter **100**, when the voltage level of VH is programmable (e.g., to 3.6V, 2.6V, or 1.8V) and when the voltage level of VL is automatically adjusted based on VH (e.g., when VH=3.6V, VL=1.8V; when VH=2.6V, VL=0.8V; and when VH=1.8V, VL=0.8V).

(35) In operation, VDDW is continuously applied to gates **142a, 142b** of NFETs **140a, 140b**, VL is continuously applied to gates **132a, 132b** of third PFETs **130a, 130b**. When IN transitions from GND to VDDW and INB transitions from VDDW to GND, in branch **101b**, gate and source voltages of NFET **140b** are at VDDW and GND, respectively, so the threshold voltage (VT) of NFET **140b** is reached. NFET **140b** turns on and pulls down the voltage on final output node **135b** to GND so OUT is at GND. When the voltage on final output node **135b** is at GND, gate voltage on first PFET **110b** is at GND and the VT of first PFET **110b** is reached. First PFET **110b** turns on, pulling down the voltage at intermediate output node **125b** to VL so OUT.sub.i is at VL. The gate and source voltages on third PFET **130b** are both at VL, so third PFET **130b** remains off. In branch **101a** the gate voltage on second PFET **120a** is at VL and the source voltage is at VH, so the VT of second PFET **120a** is reached. Second PFET **120a** turns on, thereby pulling up the voltage on intermediate output node **125a** to VH so OUTB.sub.i goes to VH. When intermediate output node **125a** is at VH, the second PFET **120b** remains off. Furthermore, the gate and source voltages on third PFET **130a** are VL and VH, respectively, so the VT of third PFET **130a** is reached. Third PFET **130a** turns on, pulling up the voltage on final output node **135a** to VH so OUTB goes to VH. Third PFET **130a** also pulls the gate voltage of first PFET **110a** up to VH, so first PFET **110a** remains off. Since the gate and source voltages of NFET **140a** are both at VDDW, NFET **140a** also remains off.

(36) When IN transitions from VDDW to GND and INB transitions from GND to VDDW, the reverse occurs. In branch **101a**, gate and source voltages on NFET **140a** are at VDDW and GND, respectively, so the VT of NFET **140a** is reached. NFET **140a** turns on and pulls down the voltage on final output node **135a** to GND so OUTB is at GND. When the voltage on final output node **135a** is GND, the gate voltage on first PFET **110a** is at GND and the VT of first PFET **110a** is reached. First PFET **110a** turns on, pulling down the voltage at intermediate output node **125a** to VL so OUTB.sub.i is at VL. The gate and source voltages of third PFET **130a** are both at VL, so third PFET **130a** remains off. Additionally, in branch **101b** the gate voltage on second PFET **120b** is at VL and the source voltage is at VH, so the VT of second PFET **120b** is reached. Second PFET **120b** turns on, thereby pulling up the voltage on intermediate output node **125b** to VH so OUT.sub.i goes to VH. When intermediate output node **125b** is at VH, second PFET **120a** remains

off. Furthermore, the gate and source voltages on third PFET **130b** are VL and VH, respectively, so the VT of third PFET **130b** is reached. Third PFET **130b** turns on, pulling up the voltage on final output node **135b** to VH so OUT goes to VH. Third PFET **130b** also pulls up the gate voltage on first PFET **110b** to VH, so first PFET **110b** remains off. Since the gate and source voltages on NFET **140b** are both at VDDW, NFET **140b** also remains off.

(37) Thus, as illustrated, OUT.sub.i at intermediate output node **125b** transitions between VL and VH and back (regardless of whether VL is at 1.8V and VH is at 3.6V or VL is at 0.8V and VH is at either 2.6V or 1.8V) and OUTB.sub.i at intermediate output node **125a** is inverted with respect to OUT.sub.i (i.e., transitions from VH to VL when OUT.sub.i transitions from VL to VH and vice versa). Additionally, OUT at final output node **135b** transitions between GND and VH and back (again regardless of whether VL is at 1.8V and VH is at 3.6V or VL is at 0.8V and VH is at either 2.6V or 1.8V) and OUTB at final output node **135a** is inverted with respect to OUT (i.e., transitions from VH to GND when OUT transitions from GND to VH and vice versa).

(38) In the disclosed circuit structure **199**, the variable voltage generator **150A** or **150B** allows the voltage level of VL to be shifted upwards (e.g., from 0.8V to 1.8V) when the voltage level of VH is relatively high, e.g., at 3.6V. Thus, all transistors **110a-110b**, **120a-120b**, **130a-130b**, **140a-140b** that are incorporated into the voltage level shifter **100** can be low voltage (e.g., 1.8V) symmetric transistors without risking operation outside the SOA for such transistors. In the example provided, the output voltage pulses would be as follows: OUT.sub.i transitions between 1.8V and 3.6V, OUTB.sub.i is inverted with respect to OUT.sub.i; OUT transitions between GND and 3.6V, and OUTB is inverted with respect to OUT. The variable voltage generator **150A** or **150B** also provides a means by which the voltage level of VL can be shifted downwards (e.g., from 1.8V to 0.8V) when, for whatever reason, a lower VH is desirable (e.g., to reduce power consumption). Those skilled in the art will recognize that in the above-described circuit structure, VH at a medium third VH level (e.g., of 2.6V) in combination with VL at a high second VL level (e.g., of 1.8V) would still leave the voltage level shifter operable; however, the switching speed of the transistors therein would be reduced. By automatically adjusting the voltage level of VL downwards (e.g., from 1.8V to 0.8V) when the medium third VH level (e.g., of 2.6V) is employed, the switching speed of the transistors is boosted, and performance is thereby improved. Those skilled in the art will also recognize that, in the above-described circuit structure, VH at a low first VH level (e.g., of 1.8V) in combination with VL at the high second VL level (e.g., of 1.8V) would leave the voltage level shifter inoperable because the threshold voltages of the transistors therein could not be reached. By automatically adjusting the voltage level of VL downward (e.g., from 1.8V to 0.8V) when VH is at the low first VH level (e.g., of 1.8 volts), the voltage level shifter remains operable.

(39) In the above-described embodiments, body biasing of the transistors within the voltage level shifter **100** can also be employed, as indicated to FIG. **1**, to improve transistor switching speeds.

(40) It should be understood that the terminology used herein is for the purpose of describing the disclosed structures and methods and is not intended to be limiting. For example, as used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Additionally, as used herein, the terms “comprises,” “comprising,” “includes” and/or “including” specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Furthermore, as used herein, terms such as “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” “upper,” “lower,” “under,” “below,” “underlying,” “over,” “overlying,” “parallel,” “perpendicular,” etc., are intended to describe relative locations as they are oriented and illustrated in the drawings (unless otherwise indicated) and terms such as “touching,” “in direct contact,” “abutting,” “directly adjacent to,” “immediately adjacent to,” etc., are intended to indicate that at least one element physically contacts another element (without other elements separating the described elements). The term “laterally” is used herein to describe the relative locations of elements and, more

particularly, to indicate that an element is positioned to the side of another element as opposed to above or below the other element, as those elements are oriented and illustrated in the drawings. For example, an element that is positioned laterally adjacent to another element will be beside the other element, an element that is positioned laterally immediately adjacent to another element will be directly beside the other element, and an element that laterally surrounds another element will be adjacent to and border the outer sidewalls of the other element. The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed.

(41) The descriptions of the various disclosed embodiments have been presented for purposes of illustration but are not intended to be exhaustive or limiting. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosed embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A structure comprising: a voltage level shifter including a low supply voltage node and a high supply voltage node; a programmable voltage generator outputting a high supply voltage to the high supply voltage node; and a variable voltage generator outputting a low supply voltage to the low supply voltage node, wherein the low supply voltage is dependent on and lower than the high supply voltage.
2. The structure of claim 1, wherein the voltage level shifter receives an input voltage pulse that transitions between an input voltage and ground and outputs multiple output voltage pulses including an intermediate output voltage pulse that transitions between the low supply voltage and the high supply voltage and a final output voltage pulse that transitions between ground and the high supply voltage.
3. The structure of claim 2, wherein the voltage level shifter further includes multiple transistors, and wherein the multiple transistors have the same voltage rating equal to the input voltage.
4. The structure of claim 3, wherein the multiple transistors are symmetric transistors.
5. The structure of claim 2, wherein the programmable voltage generator is programmable to generate and output any of a first high supply voltage equal to the input voltage and a second high supply voltage essentially double the input voltage, wherein, when the programmable voltage generator outputs the first high supply voltage, the variable voltage generator outputs a first low supply voltage lower than the input voltage, and wherein, when the programmable voltage generator outputs the second high supply voltage, the variable voltage generator outputs a second low supply voltage equal to the input voltage.
6. The structure of claim 5, wherein the programmable voltage generator is further programmable to output a third high supply voltage between the first high supply voltage and the second high supply voltage, and wherein, when the programmable voltage generator outputs the third high supply voltage, the variable voltage generator outputs the first low supply voltage.
7. The structure of claim 2, wherein the input voltage is 1.8 volts, wherein the high supply voltage is programmable to any of a 1.8-volt high supply voltage and a 3.6-volt high supply voltage, wherein, when the programmable voltage generator outputs the 1.8-volt high supply voltage, the variable voltage generator outputs a 0.8-volt low supply voltage, and wherein, when the programmable voltage generator outputs the 3.6-volt high supply voltage, the variable voltage generator outputs a 1.8-volt low supply voltage.
8. The structure of claim 7, wherein the high supply voltage is further programmable to a 2.6-volt

high supply voltage, and wherein, when the programmable voltage generator outputs the 2.6-volt high supply voltage, the variable voltage generator outputs the 0.8-volt low supply voltage.

9. The structure of claim 2, wherein the voltage level shifter includes: two input nodes receiving two input voltage pulses, respectively, wherein the two input voltage pulses include the input voltage pulse and an inverted input voltage pulse; and four output nodes outputting, in response to the two input voltage pulses, four output voltage pulses, respectively, wherein the four output voltage pulses include the final output voltage pulse, an inverted final output voltage pulse, the intermediate output voltage pulse, and an inverted intermediate output voltage pulse.

10. The structure of claim 9, wherein the voltage level shifter further includes two branches connected between the low supply voltage node and the high supply voltage node, and wherein each branch includes: an input node; an intermediate output node; an output node; a first P-type transistor; a second P-type transistor, wherein the first P-type transistor and the second P-type transistor are connected in series between the low supply voltage node and the high supply voltage node and wherein the intermediate output node is at a junction between the first P-type transistor and the second P-type transistor; a third P-type transistor having a source region connected to the junction, a gate connected to receive the low supply voltage, and a drain region connected to a gate of the first P-type transistor and to the output node; and an N-type transistor having a source region connected to the input node, a gate connected to receive the input voltage, and a drain connected to the output node, wherein, between the two branches, second P-type transistors are cross-coupled.

11. A structure comprising: a voltage level shifter including a low supply voltage node and a high supply voltage node; a programmable voltage generator receiving a trim bit signal and, based on the trim bit signal, outputting a high supply voltage to the high supply voltage node; and a variable voltage generator outputting a low supply voltage to the low supply voltage node, wherein the variable voltage generator further receives the trim bit signal and generates and outputs the low supply voltage based on the trim bit signal, and wherein the low supply voltage is lower than the high supply voltage.

12. The structure of claim 11, wherein the voltage level shifter receives multiple input voltage pulses and, in response to the multiple input voltage pulses, outputs multiple output voltage pulses including an intermediate output voltage pulse that transitions between the low supply voltage and the high supply voltage and a final output voltage pulse that transitions between ground and the high supply voltage, wherein the programmable voltage generator is programmable to generate and output any of a first high supply voltage equal to an input voltage, a second high supply voltage essentially double the input voltage, and a third high supply voltage between the first high supply voltage and the second high supply voltage, wherein, when the programmable voltage generator outputs either the first high supply voltage or the third high supply voltage, the variable voltage generator outputs a first low supply voltage lower than the input voltage, and wherein, when the programmable voltage generator outputs the second high supply voltage, the variable voltage generator outputs a second low supply voltage equal to the input voltage.

13. The structure of claim 12, wherein the variable voltage generator includes: an output voltage node connected to the low supply voltage node; a first P-type transistor with a source region connected to receive the first low supply voltage, a drain region connected to the output voltage node, and a gate connected to receive an enable signal; a second P-type transistor with a source region connected to receive the first low supply voltage, a drain region connected to the output voltage node, and a gate connected to a first output of a first NAND gate; a third P-type transistor with a source region connected to receive the second low supply voltage, a drain region connected to the output voltage node, and a gate connected to a second output of a second NAND gate; and an inverter connected to receive the trim bit signal and configured to output an inverted trim bit signal, wherein the first NAND gate is connected to receive the enable signal and the inverted trim bit signal, and wherein the second NAND gate is connected to the enable signal and the trim bit signal.

14. The structure of claim 13, wherein the enable signal controls the first P-type transistor, wherein

output signals from the first NAND gate and the second NAND gate control the second P-type transistor and the third P-type transistor, respectively, wherein, when the enable signal is at a low logic level, the first P-type transistor turns on and the second P-type transistor and the third P-type transistor turn off, so that the first low supply voltage is supplied to the low supply voltage node of the voltage level shifter through the output voltage node, wherein, when the enable signal has a high logic level and the trim bit signal has the low logic level, the second P-type transistor turns on and the first P-type transistor and the third P-type transistor turn off, so that the first low supply voltage is supplied to the low supply voltage node of the voltage level shifter through the output voltage node, and wherein, when the enable signal and the trim bit signal have the high logic level, the first P-type transistor and the second P-type transistor turn off and the third P-type transistor turns on, so that the second low supply voltage is supplied to the low supply voltage node through the output voltage node.

15. The structure of claim 12, wherein the input voltage is 1.8 volts, wherein the high supply voltage is programmable to any of a 1.8-volt high supply voltage, a 2.6-volt high supply voltage, and a 3.6-volt high supply voltage, wherein, when the programmable voltage generator outputs either the 1.8-volt high supply voltage or the 2.6-volt high supply voltage, the variable voltage generator outputs a 0.8-volt low supply voltage, and wherein, when the programmable voltage generator outputs the 3.6-volt high supply voltage, the variable voltage generator outputs a 1.8-volt low supply voltage.

16. A structure comprising: a voltage level shifter including a low supply voltage node and a high supply voltage node; a programmable voltage generator outputting a high supply voltage to the high supply voltage node; and a variable voltage generator outputting a low supply voltage to the low supply voltage node, wherein the variable voltage generator receives the high supply voltage, internally generates a trim bit signal based on the high supply voltage, and generates and outputs the low supply voltage based on the trim bit signal, and wherein the low supply voltage is lower than the high supply voltage.

17. The structure of claim 16, wherein the voltage level shifter receives an input voltage pulse that transitions between an input voltage and ground and further outputs, in response to the input voltage pulse, multiple output voltage pulses including an intermediate output voltage pulse that transitions between the low supply voltage and the high supply voltage and a final output voltage pulse that transitions between ground and the high supply voltage, wherein the high supply voltage is programmable to any of a first high supply voltage equal to the input voltage, a second high supply voltage essentially double the input voltage, and a third high supply voltage between the first high supply voltage and the second high supply voltage, wherein, when the programmable voltage generator outputs either the first high supply voltage or the third high supply voltage, the variable voltage generator outputs a first low supply voltage lower than the input voltage, and wherein, when the programmable voltage generator outputs the second high supply voltage, the variable voltage generator outputs a second low supply voltage equal to the input voltage.

18. The structure of claim 17, wherein the variable voltage generator includes: a voltage divider receiving the high supply voltage and outputting a divided high supply voltage; a comparator receiving the divided high supply voltage and a reference voltage and outputting the trim bit signal, wherein the trim bit signal is at a high logic level when the divided high supply voltage is greater than the reference voltage and at a low logic level when the divided high supply voltage is less than or equal to the reference voltage; an output voltage node connected to the low supply voltage node; a first P-type transistor with a source region connected to receive the first low supply voltage, a drain region connected to the output voltage node, and a gate connected to receive an enable signal; a second P-type transistor with a source region connected to receive the first low supply voltage, a drain region connected to the output voltage node, and a gate connected to a first output of a first NAND gate; a third P-type transistor with a source region connected to receive the second low supply voltage, a drain region connected to the output voltage node, and a gate connected to a

second output of a second NAND gate; and an inverter receiving the trim bit signal and outputting an inverted trim bit signal, wherein the first NAND gate receives the enable signal and the inverted trim bit signal, and wherein the second NAND gate receives the enable signal and the trim bit signal.

19. The structure of claim 18, wherein the enable signal controls the first P-type transistor, wherein output signals from the first NAND gate and the second NAND gate control the second P-type transistor and the third P-type transistor, respectively, wherein, when the enable signal has a low logic level, the first P-type transistor turns on and the second P-type transistor and the third P-type transistor turn off, so that the first low supply voltage is supplied to the low supply voltage node of the voltage level shifter through the output voltage node, wherein, when the enable signal has a high logic level and the trim bit signal has the low logic level, the second P-type transistor turns on and the first P-type transistor and the third P-type transistor turn off, so that the first low supply voltage is supplied to the low supply voltage node of the voltage level shifter through the output voltage node, and wherein, when the enable signal and the trim bit signal have the high logic level, the first P-type transistor and the second P-type transistor turn off and the third P-type transistor turns on, so that the second low supply voltage is supplied to the low supply voltage node through the output voltage node.

20. The structure of claim 17, wherein the input voltage is 1.8 volts, wherein the high supply voltage is programmable to any of a 1.8-volt high supply voltage, a 2.6-volt high supply voltage, and a 3.6-volt high supply voltage, wherein, when the programmable voltage generator outputs either the 1.8-volt high supply voltage or the 2.6-volt high supply voltage, the variable voltage generator outputs a 0.8-volt low supply voltage, and wherein, when the programmable voltage generator outputs the 3.6-volt high supply voltage, the variable voltage generator outputs a 1.8-volt low supply voltage.
